

KAUSHIK MASLEKAR

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<https://github.com/KaushikMaslekar> portfolio: kaushikmaslekar.vercel.app

SUMMARY

Final-year B.E. (Artificial Intelligence & Data Science) student specializing in backend engineering and distributed systems. Builds REST APIs, event-driven services, and caching layers with Java/Spring Boot and Python, and has hands-on experience with Kafka, Redis, Docker, and AWS. Additional depth in computer networks (TCP/IP, TLS, traffic analysis) and applied AI (RAG pipelines, semantic search, vector databases).

TECHNICAL SKILLS

- Backend & Distributed Systems: Java, Spring Boot, Microservices, REST APIs, Kafka, Redis, Event-Driven Architecture, Multithreading, Load Balancing
- AI/ML: Python, RAG, Semantic Retrieval, Embeddings, Pinecone, ChromaDB, Vector Search, Pandas, NumPy
- Databases: MySQL, MongoDB, PostgreSQL
- Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD
- Testing & Monitoring: JUnit, Mockito, Prometheus, Grafana, Postman

EXPERIENCE

AI/ML Intern — YBI Foundation | Feb 2026 – May 2026

- Built a Retrieval-Augmented Generation (RAG) application in Python and LangChain to enable document-grounded question answering over large document sets.
- Implemented document chunking, embedding generation, and semantic retrieval to improve response accuracy and retrieval relevance.
- Ran exploratory data analysis, statistical validation, and data cleaning to prepare datasets for ML experimentation.

KEY PROJECTS

Full-Stack RAG Chat Platform | Next.js, Spring Boot, FastAPI

- Architected a full-stack RAG chat application with a Next.js UI, a Spring Boot API gateway (JWT auth, rate limiting), and a FastAPI retrieval service communicating over SSE streaming.
- Built a ChromaDB-backed retrieval pipeline with page-aware chunking and cross-encoder reranking, achieving 83.3% recall@5 and a 0.83 mean reciprocal rank on a golden evaluation set.
- Implemented citation-grounded answer generation with automatic abstention on ungrounded queries (100% correct abstention rate), averaging 76ms end-to-end query latency and 4.2 citations per answer.

Encrypted Traffic Classifier | Java, Spring Boot, Machine Learning, Packet Analysis

- Designed an enterprise-grade encrypted traffic classification platform for network visibility and threat detection without decrypting TLS traffic.
- Built a full flow-analysis pipeline covering packet capture, flow reconstruction, and metadata/JA3 fingerprint extraction.
- Trained an ML-based traffic classifier and served it through a scalable Spring Boot backend.

MachinoCare — AI-Powered Predictive Maintenance Platform | FastAPI, IoT (ESP32), Streamlit, PostgreSQL

- Built an end-to-end predictive maintenance system for industrial machinery combining IoT sensors, ML anomaly detection, and real-time monitoring.
- Designed a FastAPI backend ingesting ESP32 vibration data over REST/WebSockets, paired with a Streamlit dashboard for live visualization and diagnostics.

EDUCATION

B.E. in Artificial Intelligence and Data Science (Pursuing)

Marathawada Mitra Mandal's College of Engineering, Pune | Aug 2024 – Present